

RELATIONAL 4-STREAM N-BACK

Product + gameplay specification

A classic N-back shell where every response requires a changing relational computation.

CORE RULE

Do not answer whether the stimulus simply repeats. Answer whether the CURRENT stimulus, under the CURRENT trial rule, maps to the stimulus N steps back.

Specification date: 31 August 2026

1. Concept

The trainer keeps the recognizable structure of N-back - a repeating sequence of trials, one shared N level, and response buttons - but replaces simple identity matching with relational matching. Each trial presents four simultaneous streams: position, spoken number, boundary shape, and an abstract relation symbol.

Design objective: Prevent the player from succeeding through rote item memorization or fixed button rules. The demanded relation changes continually, and surface-level symbol identity is deliberately made unreliable.

1.1 Trial contents

Stream	What appears	What the player compares
Position	Active cell/voxel in a 3x3 or 3x3x3 field	Does the N-back position satisfy the current spatial prompt relative to the current position?
Sound	Spoken digit 0-9	Does the N-back digit satisfy the current numeric relation (+x, -x, before, after)?
Shape	Circle/triangle/square/pentagon/hexagon framing the active location	Does the N-back shape satisfy the current side-count relation?
Relation	Abstract relational glyph inside the active shape	Does its decoded semantic relation equal the semantic relation N trials back, even when the picture differs?

1.2 Response layout

Directly below the grid/cube are four response buttons. Directly above each button is that stream's current demand. A compact example:

POSITION	SOUND	SHAPE	RELATION
ROW	-3	LESS	SAME RELATION
[Position]	[Sound]	[Shape]	[Relation]

The player presses a button only when that stream is a target for the current trial. Any combination from zero to four buttons may be correct.

2. Canonical Direction Convention

Every stream must use one convention. This specification defines the prompt as a transformation from the current stimulus to the remembered N-back stimulus:

Canonical equation: $N\text{-back stimulus} = \text{APPLY}(\text{current stimulus}, \text{current prompt}).$

This matches the intended number and shape examples and avoids one modality silently reversing the relationship.

Example	Current	Prompt	N-back required for a target
Sound	9	-3	6, because $9 - 3 = 6$
Shape	Square (side-value 4)	-3	Circle (side-value 1), because $4 - 3 = 1$
2D position	row 2, col 2	UP	row 1, col 2
2D position	row 2, col 2	ROW	any cell in row 2 (subject to generator rules)

The in-app Wiki should repeat this rule at the top of every relation family. If a different convention is later chosen, it must be changed globally rather than per stream.

3. Position Stream

3.1 2D mode - 3x3

Represent cells with fixed logical coordinates. Visual scaling must never alter the coordinates used for scoring.

Prompt	Target condition
UP	N-back cell is one row above current cell
DOWN	N-back cell is one row below current cell
LEFT	N-back cell is one column left of current cell
RIGHT	N-back cell is one column right of current cell
ROW	N-back cell shares the current row
COLUMN	N-back cell shares the current column

UP/DOWN/LEFT/RIGHT are exact one-cell relations by default. A later setting may allow ANY UP/ANY LEFT, but exact adjacency is easier to score and learn.

3.2 3D mode - 3x3x3

Use cube-local coordinates that do not change when the camera rotates. Static and rotating displays use the same trial data and same answers.

Mode	Behavior
Static cube	The cube stays at one readable oblique orientation.
Rotating cube	The cube rotates smoothly. The logical x/y/z axes remain fixed to the cube, so the player must mentally maintain orientation.
Active voxel marker	The shape frame and relation glyph are anchored to the active voxel but rendered as readable billboards so perspective does not destroy symbol recognition.

3D retains UP, DOWN, LEFT, RIGHT, ROW, and COLUMN and adds ABOVE and BELOW as the third-axis relation. The Wiki must show a cube-axis legend. Rotation should be smooth and continuous rather than random hard cuts, so difficulty comes from spatial updating rather than visual disorientation.

4. Sound / Number Stream

The auditory stimulus is a spoken digit from 0 through 9. The digit is not visually printed during normal training; a caption/accessibility option may exist, but should be clearly labeled as changing the task.

Prompt	Meaning under canonical direction
BEFORE	N-back digit is numerically lower than current digit.
AFTER	N-back digit is numerically higher than current digit.
+x	N-back digit = current digit + x.
-x	N-back digit = current digit - x.
+0	Exact numeric equality; useful as the identity case.

No wraparound is allowed. A prompt is legal only if its result stays inside 0-9. Example: for current 9, positive exact offsets greater than 0 are illegal; +0 is legal. For current 2, +7 is legal and +8 is not.

Generator requirement: Never show an impossible exact offset. For target trials, derive a valid prompt from the current/N-back pair. For non-targets, choose a valid prompt that is false for that pair.

5. Shape Stream

The shape is the boundary around the active grid cell/voxel. The interior remains visually open for the relation symbol.

Shape	App side-value
Circle	1
Triangle	3
Square	4
Pentagon	5
Hexagon	6

The circle is intentionally assigned the app-specific value 1. This is a training convention, not a mathematical claim about polygon edge count. There is no 2-sided shape, so offsets that would require value 2 are illegal.

Prompt	Target condition
MORE	N-back shape has a greater side-value than current shape.
LESS	N-back shape has a lower side-value than current shape.
+x	N-back side-value = current side-value + x, and that value exists in the shape set.
-x	N-back side-value = current side-value - x, and that value exists in the shape set.
+0	Same shape class.

Example: current Square (4), prompt -3 -> target only when the N-back shape is Circle (1). Current Triangle (3), prompt -1 is invalid because value 2 has no shape.

6. Relation Stream

The relation stream is the most important anti-memorization channel. The visible mark is not the thing to remember. The player must decode a semantic relationship, then compare that semantic relation with the relation N trials back.

Target rule: RELATION is a target when `semanticRelation(current) == semanticRelation(N-back)`, even if the two rendered glyphs are different.

The referenced GOATED Relational n-Back project uses a large relation library and a structural form-class taxonomy, including a mode that deliberately treats same-token identity differently from structural analogy. That is useful precedent for separating surface token from abstract relation. The design here should stay smaller and faster to decode during a four-stream task.

6.1 Initial semantic bank

Family	Relations to start with	Why
Directional	LEFT OF / RIGHT OF; ABOVE / BELOW	Fast, asymmetric, easy to produce visually similar variants.
Containment	INSIDE / CONTAINS	Same basic elements; relation comes from enclosure direction.
Distance	NEAR / FAR	Same two anchors; only spacing changes.
Contact	TOUCHING / SEPARATE	Same anchors; gap/contact is the relational feature.
Overlap	OVERLAP / DISJOINT	Same forms; intersection is the relational feature.
Magnitude	BIGGER / SMALLER	Same pair; relative size direction changes.
Connection	CONNECTED / UNCONNECTED	Same endpoints; only the connection state changes.
Order	BEFORE / AFTER	Two items on a path/sequence with direction cue.

Start with roughly 16 semantic relations (eight inverse pairs). Expand only after usability testing shows that users can decode them at training speed without turning the task into a symbol-language exam.

6.2 Multi-symbol requirement

- Each semantic relation must have at least 3 render variants at launch; 4-6 is preferable later.
- On an N-back relation target, use a different render variant from the remembered trial whenever possible.
- Use the same visual grammar across different relations: two anchor objects, the same stroke weights, similar sizes, and minimal unique decoration.
- Randomize nuisance properties that do not change the relation (anchor shape family, fill/hollow state, small layout jitter), but never randomize a property that accidentally changes the semantic answer.
- Generate hard non-target lures from the same renderer family as the remembered relation so visual similarity cannot be used as a shortcut.
- Prevent exact bitmap/SVG repetition inside a recent anti-repeat window unless repetition is explicitly being tested.

7. Symbol Wiki Specification

The app needs an always-accessible Wiki/Academy. It should teach semantics without encouraging memorization of one icon per relation. Each entry therefore shows several variants side-by-side.

Wiki field	Required content
Relation name	Plain label, e.g. INSIDE.
Meaning	One sentence defining A relative to B.
Inverse	The paired opposite, e.g. CONTAINS.
Variants	At least three valid visual encodings.
Near-lures	Two visually similar encodings that mean something else.
Target example	Current symbol + different N-back variant that should be matched.
Non-target example	Two similar-looking symbols that decode to different relations.

The first Wiki page should explain the global comparison direction: current -> prompt -> N-back. The relation-symbol section is the exception in that the button asks for semantic identity rather than a separately changing transform prompt.

8. Trial Generator

The generator must control semantic difficulty. Pure random generation will produce impossible prompts, accidental targets, biased relations, and exploitable frequencies.

8.1 Recommended generation order

- 1. Choose whether each stream should be a target on this trial, using balanced independent probabilities.
- 2. Choose the current stimuli: position, digit, shape, and relation semantics.
- 3. For each target stream, choose a legal prompt/semantic setup that makes the stored N-back item satisfy the rule.
- 4. For each non-target stream, choose a legal prompt that the N-back item does not satisfy.
- 5. Choose a relation render variant after semantic truth is already fixed.
- 6. Apply anti-repetition and lure constraints, then verify all four answers with a deterministic scoring function before presenting the trial.

8.2 Default balancing

Control	Recommended default
Per-stream target rate	25-30%, measured separately per stream.
Prompt repetition	No same prompt on consecutive trials when another legal prompt exists.
Relation exact-render repeat	Suppressed within N+2 trials when possible.
Multi-target trials	Allowed, but cap extreme combinations so 3-4 simultaneous targets do not dominate.
Lures	Include same-family relation-symbol lures and near-offset numeric/shape lures.
Warm-up	First N trials cannot be scored as N-back targets; still present prompts for rhythm consistency.

Critical: Generate truth first, presentation second. Scoring must never depend on what the UI happened to render.

9. Timing, Input, and Feedback

Because the task combines four channels, timing should be adjustable rather than hard-coded.

Setting	Suggested initial default
N	2
Stimulus interval	2500-3000 ms total trial window
Visual-on duration	700-1000 ms; optionally persistent grid with only active marker transient
Sound onset	Near trial onset with small fixed offset if needed for clarity
Response window	Until the next trial begins
Feedback	Optional per-trial; compact button flash or verdict after the window
Rotation speed	Slow/medium/fast presets; static is a separate mode

Keyboard bindings should mirror the left-to-right button order and be fully remappable. Input latency should be recorded, but pressing a correct stream multiple times in one trial counts once.

9.1 Feedback modes

- Off: no correctness cue during the block.
- Minimal: button briefly flashes correct/incorrect after the response window.
- Explain errors: optional training mode that names the missed relation (for example, "Sound: $9 - 3 = 6$ "). This should be disabled for serious assessment blocks.

10. Scoring and Statistics

Score each stream independently and then report a combined session view. Raw accuracy alone is not sufficient because a low target rate can reward never pressing.

Metric	Per stream?	Purpose
Hits / misses	Yes	Target detection.
False alarms / correct rejections	Yes	Controls response bias.
Balanced accuracy	Yes	Primary simple accuracy summary.
d-prime (optional)	Yes	Signal-detection measure when sample size is adequate.
Median correct RT	Yes	Speed without false-alarm contamination.
Prompt-family accuracy	Yes	Find weak relation types such as ROW/COLUMN or exact offsets.
Combined block score	Yes + overall	Summarize only after preserving stream-level detail.

If adaptive N is added, advancement should require acceptable performance on every active stream or use the weakest-stream score. Otherwise a strong easy channel could hide collapse on the relation channel.

11. Anti-Shortcut Rules

- Do not let a particular relation always use one icon, color, node shape, or orientation.
- Do not let $+x/-x$ distributions reveal the previous digit or shape through frequency biases.
- Do not wrap 0-9 arithmetic; wrapping creates a different rule family and should only exist as an explicit mode.
- Do not score 3D based on screen-space coordinates. Score cube-local coordinates.
- Do not make the rotating cube so fast that the dominant task becomes visual tracking rather than working-memory relation updating.
- Do not let exact relation-symbol repetition be the normal target cue. Semantic match with surface change should be the default.
- Do not use impossible shape offsets involving the absent side-value 2.
- Do not allow a prompt to change after stimulus onset; the rule must be available at the moment the stimulus is encoded.
- Do not reuse the same prompt for long runs. The variable rule is part of the training load.

12. Example Trial

Example at N=2. The current trial is compared with the trial two steps earlier.

Stream	Current	Prompt / rule	2-back	Correct?
Position	center cell	UP	cell directly above center	YES
Sound	9	-3	6	YES
Shape	Square (4)	-3	Circle (1)	YES
Relation	Variant C of CONTAINS	SAME RELATION	Variant A of CONTAINS	YES

On this trial all four response buttons should be pressed. Notice that the relation target is deliberately a different visual variant, so matching by picture would fail as a general strategy.

12.1 Hard non-target example

Stream	Current	Prompt	N-back	Why it is a non-target
Sound	8	-2	5	$8 - 2 = 6$, not 5.
Shape	Hexagon (6)	LESS	Pentagon (5)	This one IS a target; the generator must catch that.
Relation	LEFT OF renderer B	SAME RELATION	RIGHT OF renderer B-like	Near-identical family, different semantic relation.

This illustrates why every generated trial should be verified by the scoring engine before display: apparent non-targets can accidentally satisfy a broad relation such as LESS, ROW, or BEFORE.

13. Implementation Architecture

A clean implementation separates trial semantics from rendering. This makes the 2D/3D views interchangeable and allows the relation renderer to evolve without changing scoring.

Module	Responsibility
trialEngine	Creates trial truth, target flags, prompt legality, balancing, and anti-repeat constraints.
relationMath	Pure functions for position, digit, and shape transforms plus relation equality.
symbolLibrary	Semantic relation definitions, inverses, render-family metadata, and Wiki content.
symbolRenderer2D	Canvas/SVG rendering of relational variants.
grid2D	3x3 display and active shape/symbol composition.
cube3D	3x3x3 scene, cube-local coordinates, camera rotation, billboarded active marker.
audioEngine	Digit speech/playback with deterministic timing.
inputController	Keyboard/pointer responses and response locking.
scoring	Hit/miss/FA/CR, RTs, prompt-family statistics.
storage	Settings, session history, Wiki progress; local-first is sufficient for v1.

For the 3D implementation, Three.js is a natural fit. The referenced project also uses React/Vite, Three.js for 3D scenes, Canvas 2D for relation rendering, and client-side storage, demonstrating that this overall technical shape is practical for a browser-only trainer.

14. MVP and Expansion

MVP

- 2D 3x3 grid.
- Four streams active simultaneously.
- Position prompts: UP/DOWN/LEFT/RIGHT/ROW/COLUMN.
- Sound digits 0-9 with BEFORE/AFTER and legal +/- offsets.
- Five shape classes with MORE/LESS and legal +/- offsets.
- Eight inverse-pair relation semantics (about 16 relations), each with ≥ 3 render variants.
- Wiki with target/non-target examples.
- Per-stream scoring and session history.
- Static 3D 3x3x3 cube.

Phase 2

- Rotating 3D cube with cube-local coordinate tracking.
- More relation families after decode-speed testing.
- Lure controls and anti-shortcut difficulty settings.
- Adaptive N and adaptive trial timing.
- Practice mode that isolates one stream, then progressively combines streams.
- Optional export/import of local session data.

15. Reference Notes

Primary external reference inspected for this specification:

- GOATED Relational n-Back live app: <https://find-1-bug.github.io/goated-relational-nback-2/>
- Source repository: <https://github.com/find-1-bug/goated-relational-nback-2>

The reference project describes itself as a working-memory and relational-reasoning trainer and documents a large relation library, 3D spatial modes, client-side storage, Canvas 2D relation rendering, and Three.js scenes. Its README also describes a form-class system intended to separate abstract relational structure from exact relation-token identity. This specification borrows the general principle of surface variation while defining a different four-stream task and a much smaller, speed-oriented relation bank.

15.1 Final definition

In one sentence: A four-stream N-back in which the player continuously transforms the current position, number, and shape according to changing trial rules while separately decoding an abstract relation whose semantic identity - not its visual token - must match N trials back.